Keyu He

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Education

Carnegie Mellon University, Pittsburgh, PA — **May 2027**
Master of Science in Intelligent Information Systems @ School of Computer Science — **GPA: 4.14/4.33**

University of Southern California, Los Angeles, CA — **May 2025**
Bachelor of Science in Computer Science & Bachelor of Art in Applied Mathematics — **GPA: 3.98/4.00**
Minor in Artificial Intelligence Applications

Research Interests

Socially intelligent and multi-agent AI, evaluation of LLM systems, human–AI interaction, explainable NLP.

Skills

Programming: C++, C, Python, Java, MySQL, HTML, CSS, JS, TS, x86-64 Assembly
Frameworks/Tools/Software: PyTorch, Pandas, NumPy, LangChain, FAISS, Git, AWS, LaTeX
Areas of Expertise: Machine Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Vision Language Models (VLMs), Data Science / Data Engineering

Publications

1. **Keyu He**, Xuhui Zhou, Maarten Sap. *Social Gym and SPaRTan: Benchmarking and Improving LLM Social Reasoning via Multi-Agent Game Tournaments*. Preprint, 2026. 10.48550/arXiv.2608.09128
   [multi-agent environment engineering; LLM agent orchestration; large-scale benchmarking; self-play method design]
2. **Keyu He**, Tejas Srinivasan, Brihi Joshi, Xiang Ren, Jesse Thomason, Swabha Swayamdipta. *Believing without Seeing: Quality Scores for Contextualizing Vision-Language Model Explanations*. ACL 2026. 10.48550/arXiv.2509.25844
   [VLM metrics design; calibration; human-subject studies; statistical analyses]
3. **Keyu He***, Qianou Ma*, Valerie Chen, Wayne Chi, Tongshuang Wu. *RECAP: An End-to-End Platform for Capturing, Replaying, and Analyzing AI-Assisted Programming Interactions*. ACL Demo 2026. 10.48550/arXiv.2605.01104
   [VS Code extension; full-stack TypeScript development; cloud data pipeline design; analyses of human-AI interactions]
4. Brihi Joshi*, **Keyu He***, Sahana Ramnath, Sadra Sabouri, Kaitlyn Zhou, Souti Chattopadhyay, Swabha Swayamdipta, Xiang Ren. *ELI-Why: Evaluating the Pedagogical Utility of LLM Explanations*. Findings of ACL 2025. 10.48550/arXiv.2506.14200
   [Benchmark/Dataset design; human-subject studies; web-app implementation; multi-metric evaluation of LLM explanations]
5. Huihan Li*, Arnav Goel*, **Keyu He**, and Xiang Ren. *Attributing Culture-Conditioned Generations to Pretraining Corpora*. ICLR 2025. 10.48550/arXiv.2412.20760.
   [large-scale corpus analysis; Python data engineering and visualization; survey server infrastructure design]

Experience

Adobe, San Jose, CA — May 2026 – Aug. 2026
Machine Learning Engineer Intern

- Shipped 5+ agent skills, including a four-skill family outside the team's own domain, through an **agent skill generator** merged into the team's shared skill repo, which turns rough developer requests into standards-compliant skills, grounding against internal prior art so existing work isn't rebuilt and self-auditing against the agentskills.io spec.
- Developed the systematic quality measure for the team's agent recommendation pipeline, a **reference-free 6-dimension rubric** combining LLM-as-judge with deterministic checks, reproducible to 0.01–0.05 per-dimension variance across repeated runs and surfacing two systematic weaknesses for the team to act on.
- Took that measure to production across the org: ported it onto the internal GenAI evaluation service as a reusable scorer and staged it end-to-end for two enterprise customers, on a four-stage Databricks pipeline built to run it.

CMind, Remote May 2025 – Aug. 2025

Data Engineer Intern

- Built a LangChain-based RAG prototype over an internal analytics DB (chunking, embeddings, FAISS indexing/retrieval, LLM answer synthesis) with metadata filters and notebook/API utilities.
- Designed CEO-speech trait metrics and an auto-evaluation pipeline with custom rubrics, achieving Fleiss' κ of 0.89–0.91 against human annotations.

Teaching Experience

Carnegie Mellon University, Language Technologies Institute Aug. 2026 – Present

Graduate Teaching Assistant, 11-711 Advanced Natural Language Processing

- Supporting a graduate NLP research course spanning modeling, learning algorithms, and a paper-reimplementation project.

University of Southern California, Los Angeles, CA Sep. 2022 – May 2025

Teaching & Grading Assistant

- Coordinated logistics and grading for two CS courses (CSCI-102, CSCI-360), serving ∼300 students per term.
- Led weekly OHs/discussions for ∼20 students, clarifying concepts and providing feedback on assignments.

Projects

VLM Hallucination Mitigation — Carnegie Mellon University Feb. 2026 – Apr. 2026

- Built a training-free latent reflection module that monitors decoding entropy and injects reflection steps when the model is uncertain.
- Benchmarked against DAMO, VCD, OPERA, and DeGF on POPE/RePOPE/MMBench across LLaVA-1.5-7B, InstructBLIP-7B, and Qwen2-VL-7B.
- On RePOPE, raised LLaVA-1.5-7B accuracy from 84.0% to 90.3% with the lowest calibration error (ECE 0.028) among all methods compared.

LLM Prompt Recovery Project — University of Southern California Mar. 2024 – Apr. 2024

- Built a system to reconstruct user prompts from original text and Gemma-modified outputs.
- Fine-tuned Mixtral-8x7B with custom similarity metrics (sentence-T5-base + sharpened cosine), achieving a 0.65 leaderboard score.
- Earned a Kaggle silver medal and released the final fine-tuned model publicly on Kaggle.

Major Awards

- **USC CURVE Research Fellowship**, awarded in 4 terms (S24, Su24, F24, S25); total funding **$6,750**
- **USC Academic Achievement Award**, awarded in 4 terms (F22, S23, S24, F24); total support ≈**$24,000**
- **Silver Medal**, Kaggle Competition, Ranked 75/2175 (Top 3.4%) on the global leaderboard, LLM-Prompt-Recovery Project, 2024
- **1st Prize**, International Linguistics Olympiad (Senior Level), Individual Open Round, China, 2021
- **1st Prize**, International Linguistics Olympiad (Senior Level), Team Open Round, China, 2021
- **4th Place**, USC Integral Bee Competition, 2022